

Optimal Control Methods for Compensating Packet Losses in Networked Control Systems over Markov Channels

Ricardo A. Alayon M.¹, Andres L. Jutinico A.¹, and

¹ Universidad Distrital Francisco José de Caldas, Bogotá, Colombia
raalayonm@udistrital.edu.co

² Universidad Distrital Francisco José de Caldas, Bogotá, Colombia
aljutinicoa@udistrital.edu.co

Abstract. With the advancement of the Internet of Things (IoT), it is desirable to develop control systems that include communication loops in feedback loops. For example, wireless networked control systems are particularly relevant in robotic assistance and rehabilitation systems, precision agriculture, space exploration, and cooperative robotics, among others. However, wireless networked control systems are sensitive to communication problems such as delays, packet loss, and bandwidth limitations. Among these, packet loss can degrade the performance of the closed-loop system and affect its stability. Therefore, this paper introduces optimal control methods to compensate for packet loss in a communication channel. The channel is modeled as a finite-state Markov chain that captures the stochastic nature of packet loss. Two controllers are tested: Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC), both aimed at reducing losses and managing control effort. The choice of these two techniques is based on their shared characteristics of optimizability and closed-loop robustness. However, our proposal focuses on the predictive control action capability offered by MPC, which allows a set of control actions to be sent in each packet, enabling the controller to mitigate the effects of packet loss. Simulations were performed to observe the effect of lost packages on the control action and state variables of a dynamical system, using RMS values to quantify performance. The results show that LQR is sensitive to packet loss, even decreasing control effort and resulting in degraded speed response. MPC, on the other hand, maintains stability by gradually increasing control actions, demonstrating a better trade-off between performance and control effort. These findings suggest that predictive control is more robust to typical network uncertainties, at the cost of increasing control actions, but providing consistent performance while managing control energy more effectively.

Keywords: Linear Quadratic Regulator(LQR) · Model Predictive Control(MPC) · Control over Wireless Networks · Finite-state Markov Chain

1 Introduction

Networked control systems (NCS) have become increasingly relevant as wireless communications become widespread in modern engineering applications. Systems such as autonomous vehicles, mobile robots, sensor networks, and unmanned aerial vehicles rely on wireless links to exchange and process information between sensors, controllers, actuators, and even human operators. Although replacing wired communication with wireless technologies enables greater mobility and scalability, it also introduces significant challenges related to reliability and performance. In particular, wireless communication channels are inherently affected by stochastic phenomena such as packet losses and time-varying connectivity. These effects degrade closed-loop performance, compromise stability, and reduce control accuracy in feedback systems. As a consequence, the design of control strategies must explicitly account for communication uncertainties. A common model approach to these uncertainties is through Markov communication channels, which describe the random evolution of packet transmission success or failure over time. In this context, the interaction between the controller and the plant is influenced by the probabilistic nature of the communication medium, leading to hybrid system behavior.

Several optimal control strategies have been proposed in the literature, including the Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC). Although these methods are well-established in deterministic scenarios, their performance under stochastic communication constraints remains an active research topic. In this context, this paper presents a comparative analysis of the LQR and MPC applied to a closed-loop system with a Markovian communication channel. The study evaluates the impact of channel-induced packet losses on closed-loop performance and the ability of each control strategy to mitigate them. Transmission delays and external disturbances are intentionally excluded in order to isolate the impact of the Markovian packet-loss mechanism.

2 Related Work

Networked control systems (NCS) have been studied under communication constraints, including delays, packet dropouts, and channel uncertainties. Early works on discrete-time Markovian jump linear systems provide an important framework for modeling stochastic switching behaviors induced by networked environments [6]. In this context, state delays or parametric uncertainties have been considered primary sources of performance degradation, as in [1, 3], where robust regulation strategies are developed for systems subject to random delays and polytopic uncertainty descriptions. More recently, communication-aware control strategies have incorporated explicit models of wireless channels. In particular, Markovian channel models have been adopted to capture time-varying network reliability and packet dropout phenomena. Works such as [2, 15] address optimal and robust control design over wireless channels, with mechanisms to compensate for packet loss, and emphasize the impact of stochastic communication on

closed-loop performance. Similarly, Lyapunov-based approaches for Markovian switching systems with network-induced uncertainties have been proposed in [4, 17], providing stability guarantees in the mean-square sense.

In terms of estimation and filtering, recursive and Kalman-based approaches have been proposed to accommodate Markovian jump dynamics and random delays [10, 11]. These methods have focused on the responsibility of uncertainty propagation through the communication network and its effect on state reconstruction accuracy. In addition, secure and reliable state estimation over Markov channels has been investigated in [13], where communication constraints are explicitly incorporated into the estimation process. Control over systems conditioned by communication has also been addressed from an optimization perspective. In particular, linear quadratic regulation under network restrictions has attracted the researchers' attention, with results highlighting notorious performance limitations in learning-based and optimal control frameworks [12]. Extensions to more complex dynamical structures, including quadrotor systems under external disturbances, further illustrate the relevance of robust and intelligent control design in uncertain environments [5]. Despite these advances, the interaction between classical optimal control techniques and communication channels modeled by Markov processes remains an active area of research. Namely, the combined effect of controller structure and channel-induced stochasticity continues to pose challenges ranging from correctly understanding and modeling the structure of uncertainties to analyzing and predicting the performance degradation caused by these communication uncertainties, and ultimately ensuring system performance, stability, and robustness.

3 Methodology

3.1 Model Formulation

Consider a discrete-time closed-loop control scheme, as shown in Fig 1, where the feedback loop is implemented over a wireless communication channel, which can be modeled as a Markov process. In this context, the feedback signal is not transmitted reliably, due to packet loss, which directly affects the performance and stability of the closed loop. To analyze this effect, the following discrete-time model is considered:

$$x_{k+1} = Fx_k + Gu_k, \quad (1)$$

where $x_k \in \mathbb{R}^n$ represents the state vector, u_k is the control input, and F, G are the discrete-time representation of the matrices A and B of the dynamic system shown in Fig 1. When the feedback loop is closed through a wireless channel, the control input is affected by a binary stochastic process $S_k \in \{0, 1\}$, where $S_k = 1$ represents packet loss and $S_k = 0$ represents successful transmission during each sample time T_s , respectively. Under this consideration we define the following hybrid model that considers the dynamic system and the communication channel:

$$x_{k+1} = Fx_k + G(1 - S_k)u_k. \quad (2)$$

The evolution of S_k is modeled as a Markov process, usually employed for representing bursts in packet loss behavior, as usually observed in wireless communication channels due to the persistence of bad channel conditions. Figure 1 shows the resulting networked control architecture, where the controller, process plant, and communication channel form a hybrid feedback system.

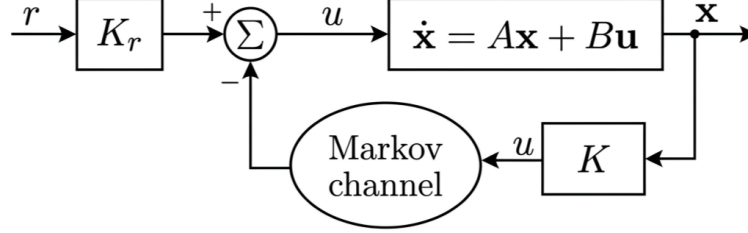


Fig. 1: Block diagram for the hybrid system

Markovian Communication Channel The feedback signal is transmitted through a Markovian channel inspired by the Gilbert–Elliot model. The binary process S_k governs the successful or failed transmission of control actions. The equivalent channel model could be expressed as:

$$y_k = x_k \cdot (1 - e_k) \quad (3)$$

Where e_k comes from the Markov state process $S_k \in \{0, 1\}$ when 0 represents the good channel state, 1 the bad state and the inner dynamics are ruled by:

$$P(S_{k+1} = j | S_k = i) = p_{ij}, \quad (4)$$

with the following probability transition matrix,

$$P = \begin{bmatrix} p_{GG} & p_{GB} \\ p_{BG} & p_{BB} \end{bmatrix}. \quad (5)$$

The above allows evaluating different network conditions by varying the probability of remaining in the good channel state. Notice that: $(P_{GB} = x) \rightarrow (P_{GG} = 1 - x)$ what implies that, it is equivalent to say 80% probability of stay in the good channel state (corresponding to P_{GG}), than express it as switch to bad state probability 20% (corresponding to P_{GB}). In this investigation, we considered the probabilities of switching from a good channel to a bad channel as follows:

$$P_{GB} \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3\}. \quad (6)$$

It is also necessary to note that there are state-dependent probabilities of communication error. Typically, close to zero in the good state and close to 1 in the

bad state. Therefore, a state switch to bad implies more frequent error occurrence. The good state has a near-zero error probability. However, errors can still occur in it. Figure 2 illustrates how distinct transition probabilities affect the occurrence of errors and their length. As a consequence of the Markov process, there are bursts of packet losses. The Markov chain induces temporal correlation in the states, then the model captures bursty errors which are related directly with the channel parameters.

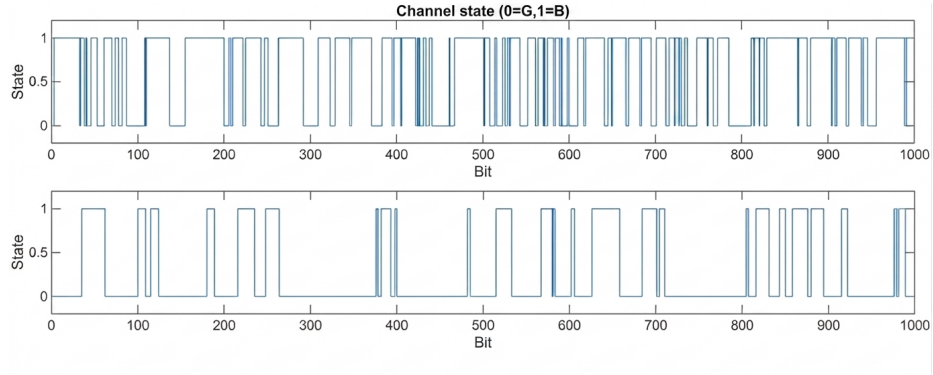


Fig. 2: Channel with two state transition probabilities. Top: 85% probability of Good-to-Bad transition. Bottom: 96% probability of good-to-bad transition.

3.2 Control Strategies

The foundations of optimal control theory are closely linked to the works of Bellman on dynamic programming [20] and Pontryagin's maximum principle [19], which established the core mathematical framework for modern optimal control. These ideas were later formalized in the context of linear-quadratic regulation, particularly through Kalman's contributions [21] and the comprehensive treatment by Anderson and Moore [22]. Together, these works define the classical LQR framework that remains widely used in state-feedback design. Model Predictive Control (MPC) originated from early industrial and heuristic approaches to constrained control [23, 24]. Its later theoretical development provided rigorous guarantees in terms of stability and optimality, as established in the work of Rawlings and Mayne [25] and Mayne et al. [26]. This progression has made MPC a standard methodology for constrained multivariable systems, particularly in applications where prediction and constraint handling are essential.

Readers already familiar with control theory will recognize these results as standard elements of the field, while those less familiar are referred to the cited works for a more complete and rigorous treatment of the underlying concepts. The controller designs adopted in this study follow these well-established formulations, which are used here without further modification due to their maturity in

the literature. Based on this framework, the following sections present the simulation setup and a detailed analysis of the obtained results under the considered communication constraints.

3.3 Model and Parameters for Control Design

A DC motor system was employed as the plant. The motor states considered are angular velocity (ω) and current (i) and the control action is the voltage u . The continuous-time dynamics are ruled by:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad A = \begin{bmatrix} -\frac{R}{L} & -\frac{K_e}{L} \\ -\frac{K_t}{J} & -\frac{b}{J} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}. \quad (7)$$

The model parameters were based on the Quanser QUBE DC motor ([18]). Table 1, exhibit a complete list of physical, computational and required parameters necessary to successfully perform simulations.

Table 1: Complete Simulation and Controller Parameters

Category	Parameter / Description	Value
DC Motor (QUBE)	Armature resistance (R)	6.3 Ω
	Armature inductance (L)	0.00085 H
	Back-emf constant (k_m)	0.036 V · s/rad
	Rotor inertia (J)	4.0×10^{-6} kg · m ²
	Viscous damping (b)	5.7×10^{-4} N · m · s/rad
Open-Loop Test	Step input voltage (u)	4.0 V
Simulation Setup	Sampling period (T_s)	0.001 s
	Simulation time window (t_{fin})	0.1 s
	Samples (N)	2000
	Reference speed (r)	3.0 rad/s
	Initial state conditions ($\mathbf{x}_0$)	$[0 \ 0]^T$
MPC Controller	Control horizon (m)	3
	Prediction horizon (p)	10
	Actuator constraints (MV)	$[-\infty, \infty]$ V
LQR	Weighting matrices	$Q = \begin{bmatrix} 50000 & 0 \\ 0 & 7 \end{bmatrix}, R=1$

It is worth noting that the selection of LQR and MPC is based on the fact that both are model-based controllers and share characteristics in the definition of control loop vectors when MPC does not consider constraints or state estimation. Table 1 shows the design parameters for both MPC and LQR. The MPC was designed with a control horizon of 3 sampling periods and a prediction horizon of 10 samples. The LQR was designed with the Q and R weighting matrices shown in the table. The difference in the operation of these control strategies

in the packet-loss system arises because the MPC can send the current control action and predictions of the next two control actions in each packet. In contrast, the LQR can use only the current control action. The above allows the MPC-controlled control system to recover from lost packets through prediction, up to two consecutive sampling periods, with only a small memory buffer required.

3.4 Performance Evaluation

The performance of each control strategy is evaluated under identical initial conditions and reference trajectories, using the same control law in each case. Each simulation generates current, velocity, and control input trajectories, which are used to assess system performance.

This illustrates two aspects: the degradation caused by communication channel errors, and the distinct behavior of each control method. Dashed red traces in each plot show the ideal (loss-free) performance, for comparison with closed-loop simulations over the imperfect wireless channel.

Root mean square (RMS) values of current, velocity, and control input are computed for comparative analysis under different packet loss probabilities, providing a single quantitative measure of performance.

The figures illustrate the effects of the communication channel during the first 100 ms, using the same sequence of channel errors with identical probabilities. This short interval highlights transient behavior. Simulations and RMS calculations are performed over 2 seconds. As predicted analytically, closing the loop over the wireless link introduces noisy behavior with LQR. The system remains stable, but velocity is particularly affected by information loss. When errors persist, the trajectory is visibly far from the reference (Figure 3), which represents the worst-case scenario considered.

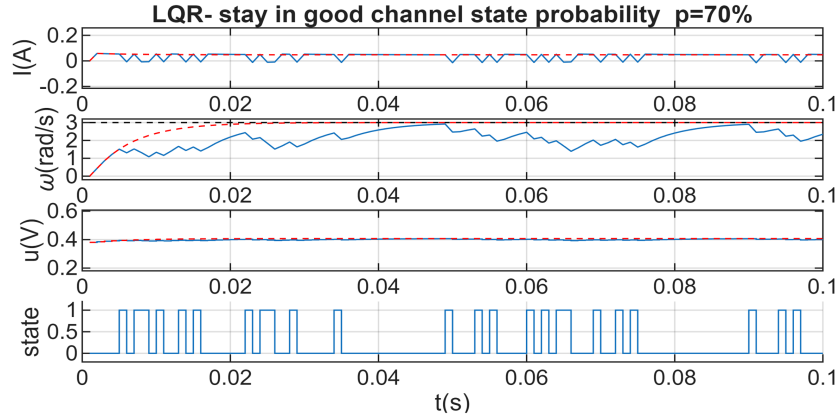


Fig. 3: System evolution for LQR controller with probability ($P = 70\%$).

MPC (Figure 4) stabilizes quickly with minimal oscillations. The probability corresponds to the worst-case scenario studied. At first glance, predictive control

reduces the noisy behavior seen with LQR, showing its robustness under channel errors.

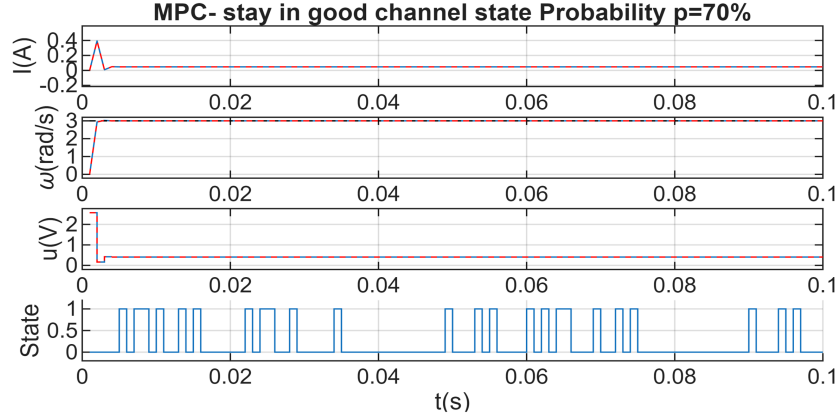


Fig. 4: System evolution for MPC controller with probability ($P = 70\%$).

4 Results

The hybrid system, simulated by closing the control loop over a Markov channel, exhibits degraded behavior compared to the "wired" system, due to transmission errors induced by the stochastic channel. Trajectories vary with different error probabilities, showing the impact of channel errors on the control signals.

For LQR, clear differences can be seen in current and velocity. The bottom curves, corresponding to a 15% Good-to-Bad state transition, appear noisy (Figure 5). LQR shows some susceptibility to channel variations, with noticeable oscillations and deviations from reference values that increase as errors persist.

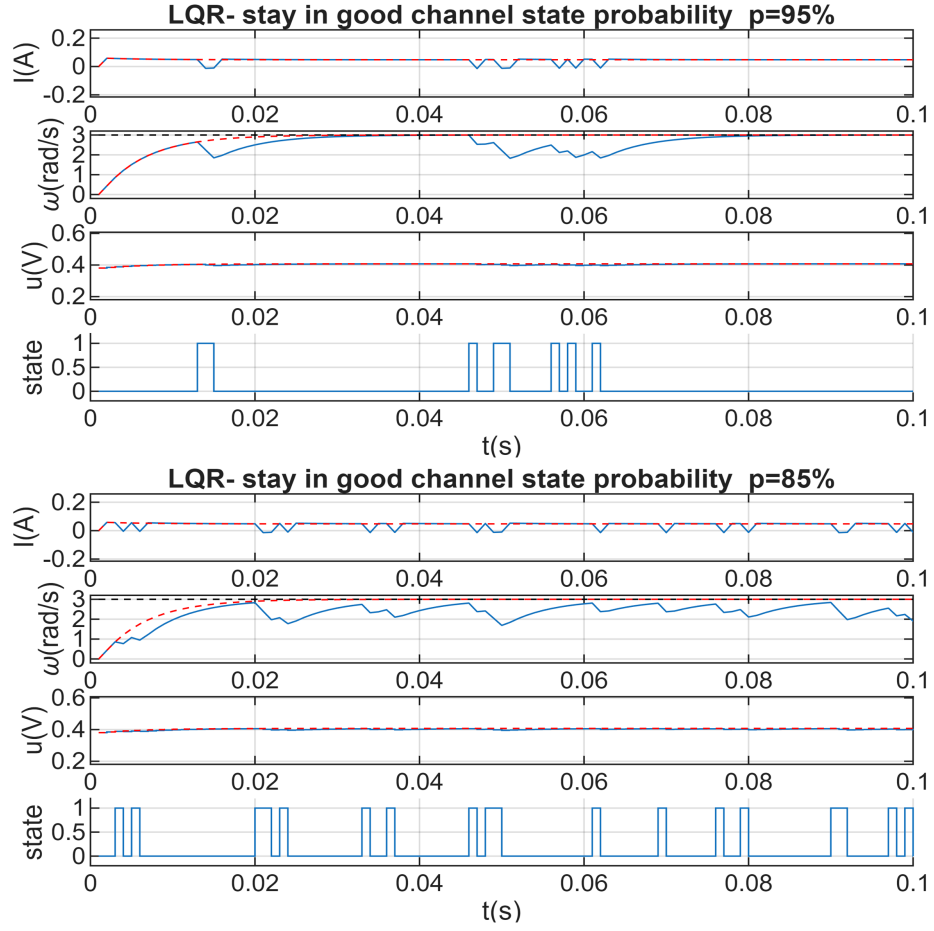


Fig. 5: Impact on the LQR closed-loop system. Top: 5% channel transition probability curves. Bottom: 15% channel transition probability curves, illustrating performance degradation.

MPC shows clear improvement, with control signals quickly stabilizing and minimal oscillations or noise (Figure 6). The traces remain almost identical across the observed time window, demonstrating MPC's robustness under different channel probabilities. Longer intervals could show more noise, which RMS measurements can confirm.

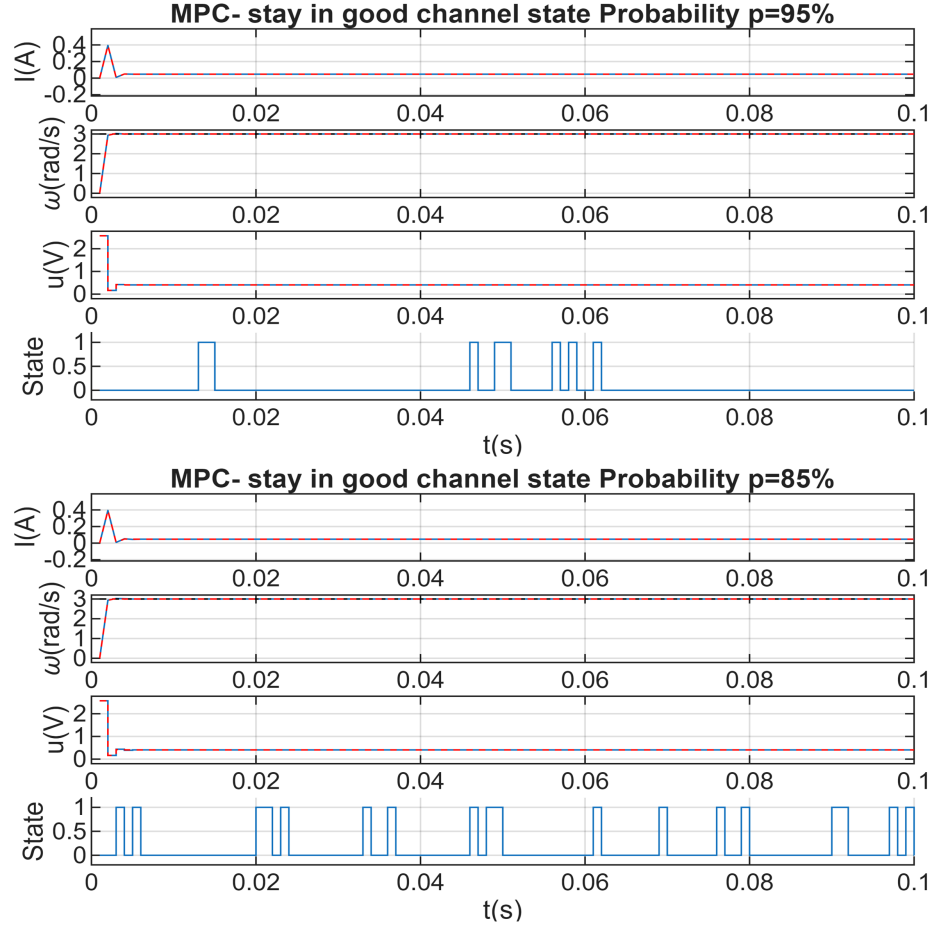


Fig. 6: Impact on the MPC closed-loop system. Top: 5% channel transition probability curves. Bottom: 15% channel transition probability curves, illustrating performance degradation.

Based on this analysis, LQR performs the worst, and MPC performs the best, with MPC providing more robust and accurate tracking of the reference values. RMS analysis provides quantitative evidence supporting this comparative evaluation.

This section is devoted to presenting and analyze the numerical measurements suggested during the performance evaluation section, in order to confirm or revise the first sight *a priori* conclusions reached when examining the plots presented in the previous section, in addition to the "*worst cases*" shown in section 3 .

The RMS analysis begins by computing these measures in simulation under all possible channel uncertainty values. The results are systematically presented in the following tables, along with their corresponding analyses. Table 2 exhibits the RMS values for current, velocity and control response under every probability value for the Markov channel performing control with LQR. For each variable, two columns are provided: "RMS" and "Delta". "RMS" shows the numerical root mean square measured from the signal under a lossy channel while "Delta" represents its difference from the RMS calculated over a "wired" (lossless) channel taken as nominal value.

Table 2: LQR Performance Measures Under Markovian Packet Dropout

Channel Reliability	Current I (A)		Speed ω (rad/s)		Control u (V)	
	RMS	Delta	RMS	Delta	RMS	Delta
Ideal (100%)	0.047523	–	2.993309	–	0.407171	–
99%	0.047424	-0.000099	2.964727	-0.028582	0.406903	-0.000268
98%	0.047276	-0.000247	2.924353	-0.068956	0.406524	-0.000647
97%	0.047129	-0.000394	2.881188	-0.112121	0.406129	-0.001042
96%	0.046879	-0.000644	2.818399	-0.174910	0.405541	-0.001630
95%	0.046892	-0.000631	2.823514	-0.169795	0.405584	-0.001587
90%	0.046118	-0.001405	2.648143	-0.345166	0.403948	-0.003223
85%	0.045722	-0.001801	2.553493	-0.439816	0.403105	-0.004066
80%	0.044763	-0.002760	2.369091	-0.624218	0.401423	-0.005748
75%	0.044328	-0.003195	2.298874	-0.694435	0.400759	-0.006412
70%	0.043946	-0.003577	2.224872	-0.768437	0.400125	-0.007046

LQR shows a progressive reduction in speed performance as channel reliability decreases. From 100% to 70% reliability, the speed RMS drops noticeably, while both current and control RMS values decrease only slightly. This behavior indicates that the controller becomes less active under packet losses, leading to lower control effort but also reduced tracking capability. In particular, the control signal u shows the highest sensitivity to packet losses, reflecting an increase in the control effort needed to compensate for the uncertainty of the channel. The state variables, current i and angular speed ω , show more moderate variations, although they still present a consistent trend of degraded tracking performance.

The state variables, current i and angular velocity ω , present a more stable response than in the LQR case, with smaller relative variations in RMS across the different loss scenarios. This suggests a better capacity to compensate for the uncertainty introduced by the communication channel.

Table 3 shows the same RMS measurements from simulation performed with the MPC controller.

Table 3: Model Predictive Control (MPC) Performance Measures Under Markovian Packet Dropout

Channel Reliability	Current I (A)		Speed ω (rad/s)		Control u (V)	
	RMS	Delta	RMS	Delta	RMS	Delta
Ideal (100%)	0.049027	–	2.994560	–	0.414337	–
99%	0.049027	0.000000	2.994560	0.000000	0.414337	0.000000
98%	0.050120	0.001093	2.987444	-0.007116	0.419309	0.004972
97%	0.051121	0.002094	2.981026	-0.013534	0.423826	0.009489
96%	0.052908	0.003881	2.968349	-0.026211	0.431940	0.017603
95%	0.052709	0.003682	2.969731	-0.024829	0.430929	0.016592
90%	0.057067	0.008040	2.937240	-0.057320	0.451300	0.036963
85%	0.058951	0.009924	2.923461	-0.071099	0.460332	0.045995
80%	0.062939	0.013912	2.890254	-0.104306	0.479684	0.065347
75%	0.064008	0.014981	2.881469	-0.113091	0.484986	0.070649
70%	0.065326	0.016299	2.871238	-0.123322	0.491590	0.077253

The MPC controller maintains the speed response close to the nominal condition, even under severe packet dropout. However, this robustness is accompanied by an increase in both current RMS and control RMS as channel reliability decreases, indicating a higher control effort under degraded network conditions.

The state variables, current i and angular velocity ω , show generally consistent behavior, with limited degradation in tracking performance. In contrast, the control signal u presents a clear increase in magnitude variation as packet loss increases. This behavior is consistent with the predictive nature of MPC, which adjusts the control action based on the expected system evolution.

Although the main objective is speed control, current variations are also considered to provide a more complete view of the system behavior. These variations reflect the coupling between the control input (u) and the system dynamics, and help to better interpret the overall response under communication degradation.

Overall, the results indicate that MPC is the most robust controller among those evaluated, maintaining an almost constant velocity RMS performance even under severe channel degradation, at the cost of a higher control effort.

5 Discussion

The comparative RMS analysis under different packet loss levels reveals clear differences between the evaluated controllers. The results show that the MPC controller achieves better velocity tracking compared to LQR, especially as channel reliability decreases. This improvement is also visible in Figure 7.

In terms of performance, the velocity RMS deviation for MPC remains low, ranging from 0 to 0.123. In contrast, the LQR controller shows a much larger variation, reaching up to 0.768. These results indicate that LQR is more sensitive to packet loss probability, and its tracking performance degrades more significantly under poor communication conditions.

Regarding control effort, MPC shows an increase in both current and control RMS as channel reliability decreases, indicating a higher effort to maintain

performance. On the other hand, LQR shows a slight reduction in control effort, which is related to a loss of control effectiveness rather than efficiency.

The LQR controller does not actively adjust its behavior under packet losses and exhibits a more conservative response in degraded conditions. As a result, both current and speed performance degrade more compared to MPC.

Figure 7 shows the RMS values for each state variable under different packet loss probabilities. The plots highlight that MPC maintains more stable speed performance across all scenarios, while LQR exhibits stronger degradation as reliability decreases. In contrast to LQR, MPC continuously adjusts its control action, improving performance under uncertainty.

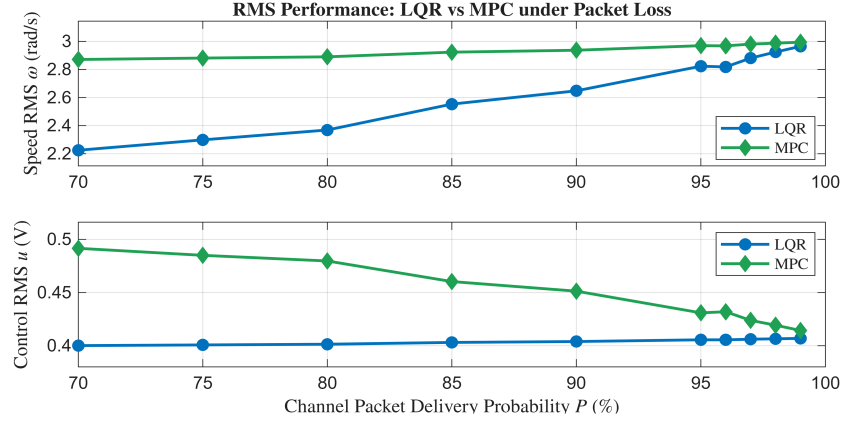


Fig. 7: Performance comparison under packet loss probabilities

Overall, the results indicate that LQR is highly sensitive to communication failures, while MPC provides more robust and consistent performance under degraded channel conditions, at the cost of a higher control effort.

6 Conclusion

In this investigation, we studied two optimal control methods, MPC and LQR, to compensate for packet loss in a communication channel. To ensure a fair comparison, the experiments considered the same dynamic system (a DC motor) and the same channel conditions, modeled as a finite-state Markov chain. We performed simulation-level experiments using MPC with a three-period control horizon, where each packet transmits the current control action and the next two predictions. The MPC maintained speed tracking close to nominal behavior at all reliability levels, even under severe packet loss. This robustness was accompanied by increased RMS values of the control signal and current, indicating that the MPC requires greater control effort to maintain performance. The LQR maintained stability across all tested conditions despite only transmitting the current control action per packet. However, its speed performance was significantly affected by packet loss. As channel reliability decreases, tracking performance progressively degrades, making LQR less suitable for networked

control systems operating with unreliable communication channels. The results also show that LQR reduces its control effort as the channel degrades, reflected in a decrease in the RMS of the control signal. Overall, the results indicate the advantages of predictive control strategies in networked control systems subject to stochastic packet loss. The RMS analysis shows a clear trade-off between performance and control effort, where MPC achieves a higher rate-tracking performance at the cost of greater actuation compared to LQR. The study also shows that the controller's behavior in response to packet loss is linked to its handling of feedback degradation.

References

1. Odorico, E.K., Terra, M.H.: Robust regulation of discrete-time systems subject to parameter uncertainties and state delay. *Automatica* **156**, 111179 (2023). doi:10.1016/j.automatica.2023.111179
2. Zacchia Lun, Y., Smarra, F., D'Innocenzo, A.: Optimal control over Markovian wireless communication channels under generalized packet dropout compensation. arXiv preprint arXiv:2501.17105 (2025). doi:10.48550/arXiv.2501.17105
3. Odorico, E.K., Bueno, J.N.A.D., Terra, M.H.: Robust recursive regulator for polytopic Markovian jump linear systems with random state delays. *IEEE Transactions on Automatic Control* (2024). doi:10.1109/TAC.2024.3423568
4. Impicciatore, A., Pepe, P., D'Innocenzo, A.: Lyapunov characterizations of exponential mean square input-to-state stability for discrete-time Markovian switching nonlinear systems. *IEEE Transactions on Automatic Control* **68**(8), 4706–4721 (2023). doi:10.1109/TAC.2022.3211777
5. Simplício, P., Benevides, J., Inoue, R., Terra, M.H.: Robust and intelligent control of quadrotors subject to wind gusts. *IET Control Theory & Applications* **18**, 2594–2611 (2024). doi:10.1049/cth2.12615
6. Costa, O.L.V., Fragoso, D.M., Marques, R.P.: *Discrete-Time Markovian Jump Linear Systems*. Springer, New York (2005)
7. Sadeghi, K., Rapajic, P.B., Shams, R.: Finite-state Markov channel modeling. *IEEE Wireless Communications* **15**(3), 7–13 (2008).
8. Araniti, G., Campolo, C., Condoluci, M., Iera, A., Molinaro, A.: LTE for vehicular networking: a survey. *IEEE Communications Magazine* **51**, 148–157 (2013). doi:10.1109/MCOM.2013.6515060
9. Dai, H., Bai, Z.: On eigenvalue bounds and iteration methods for algebraic Riccati equations. *Journal of Computational Mathematics* **29** (2011). doi:10.4208/jcm.1010-m3258
10. Odorico, E.K., Rocha, K.T., Terra, M.H.: Recursive estimation for discrete-time Markovian jump singular systems with random state delays. *IFAC-PapersOnLine* **52**(2), 4162–4167 (2019). doi:10.1016/j.ifacol.2019.12.225
11. Rocha, K.T., Terra, M.H.: Robust Kalman filter for systems with parametric uncertainties. *Systems & Control Letters* **157**, 105034 (2021). doi:10.1016/j.sysconle.2021.105034
12. Lee, B., Ziemann, I., Tsiamis, A., Sandberg, H., Matni, N.: The fundamental limitations of learning linear-quadratic regulators. In: *Proc. IEEE Conference on Decision and Control (CDC)*, pp. 4053–4060 (2023). doi:10.1109/CDC49753.2023.10383608

13. Impicciatore, A., Zacchia Lun, Y., Pappas, G.: Secure state estimation over Markov wireless communication channels. In: *Proc. IEEE CDC*, pp. 2935–2940 (2022). doi:10.1109/CDC51059.2022.9992668
14. Impicciatore, A., Grifa, M., Pepe, P., D’Innocenzo, A.: Sufficient Lyapunov conditions for exponential mean square stability of discrete-time systems with Markovian delays. In: *Proc. IEEE Mediterranean Conference on Control and Automation (MED)*, pp. 1305–1310 (2021). doi:10.1109/MED51440.2021.9480250
15. Zacchia Lun, Y., Santucci, F., D’Innocenzo, A.: Robust linear quadratic regulation over Markov channels under packet dropout compensation. *IEEE Control Systems Letters* (2024). doi:10.1109/LCSYS.2024.3521656
16. Meskin, N., Khorasani, K.: Fault detection and isolation of Markov jump linear systems. *Automatica* **45**, 2032–2040 (2009). doi:10.1016/j.automatica.2009.04.020
17. Impicciatore, A., Pepe, P., D’Innocenzo, A.: Lyapunov characterizations of exponential mean square ISS for Markovian systems. *IEEE Transactions on Automatic Control* (2023). doi:10.1109/TAC.2023.3328268
18. Quanser Inc.: QUBE-Servo DC Motor Modeling. Technical Manual, Quanser Inc., Markham, ON, Canada (2014)
19. Pontryagin, L.S., Boltyanskii, V.G., Gamkrelidze, R.V., Mishchenko, E.F.: *The Mathematical Theory of Optimal Processes*. Interscience, New York (1962)
20. Bellman, R.: *Dynamic Programming*. Princeton University Press, Princeton, NJ (1957)
21. Kalman, R.E.: Contributions to the theory of optimal control. *Bol. Soc. Mat. Mexicana* **5**, 102–119 (1960)
22. Anderson, B.D.O., Moore, J.B.: *Optimal Control: Linear Quadratic Methods*. Prentice Hall, 1989
23. Richalet, J., Rault, A., Testud, J.L., Papon, J.: Model predictive heuristic control: Applications to industrial processes. *Automatica* **14**(5), 413–428 (1978)
24. Cutler, C.R., Ramaker, B.L.: Dynamic matrix control—a computer control algorithm. *Proceedings of the Joint Automatic Control Conference*, pp. 17–23 (1980)
25. Rawlings, J.B., Mayne, D.Q.: *Model Predictive Control: Theory and Design*. Nob Hill Publishing (2009)
26. Mayne, D.Q., Rawlings, J.B., Rao, C.V., Scokaert, P.O.M.: Constrained model predictive control: Stability and optimality. *Automatica* **36**(6), 789–814 (2000)